How UT CMP is different from past University application platforms

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Introduction

While initially designed as a replacement for PyPE and DEM platforms, UT CMP is a multi-language application platform, supporting Python, Java, PHP, and other languages as identified and requested by the UT developer community, and approved by the Information Security Office (ISO). The platform was designed with security as its ***highest*** priority. Developer experience, including incorporating the twelve-factor app methodology, is a close second.

What does all this mean? In sum, many of the practices teams had followed with other platforms can no longer be used.

Generic base images

UT CMP is a multi-language platform and, as such, any base image that UT CMP supplies to developers will be more generic than what was provided in PyPE. This shift means that Python app developers have the control and responsibility of importing and updating libraries that previously were bundled automatically with PyPE toolsets.

For more information, refer to Base images.

UT CMP's CI/CD pipeline with UT Austin GitHub

Learn more about the UT CMP Deployment Model.

Application development methodology

Developers now have more control to include only packages that they use, and it opens the door for teams to make changes such as easily using Flask instead of Django for their web framework, or upgrading their Django version without having to wait for the central team to update the base image.

This design was in part influenced by the **Twelve-Factor App methodology**, a set of best practices designed to enable applications to be built with portability and resilience when deployed to the web. A twelve-factor app does not rely on the implicit existence of system-wide packages; instead, it declares all dependencies, completely and exactly, for explicit control of the code being developed. This methodology was, in part, followed because the Information Security Office (ISO) recommends that no code is deployed that includes unused dependencies. When an application is deployed with code you are not aware of and do not use, it creates security vulnerabilities. By removing all packages except what is explicitly required in UT CMP images, developers have more control over what they deploy and are responsible only for what they choose to include.

Secure deployments required

One of the advantages of containerized deployments is that developers have more control over the process of building an image containing their code. This allows the application platform to take an agnostic approach to deployments, and gives development teams more options in terms of tools, practices, etc. In essence, developers build a valid image and the infrastructure runs that image, which is a divergence from platforms like PyPE.

While this approach allows developers much latitude, it also means there needs to be processes in place to ensure deployments are free of security vulnerabilities. During the build and deployment process, UT CMP scans images for security vulnerabilities and will deny deployments that exceed ISO security thresholds. This means development teams need to update dependencies and base images to safe versions - and they must do this more frequently than they might have in the past.

The UT CMP team is developing processes to actively scan images that are used in current deployments. They will notify development groups of vulnerabilities and give a grace period to amend the code.

As the service matures, so will these processes and checks.

Search UT CMP Documentation

Search

Service Links

- UT CMP Rancher
- UT CMP Harbor
- UT GitHub
- UT Backstage
- UT Splunk CMP Dashboard
- Request to Onboard GitHub Org to UT CMP
- Subscribe to utcmp-users

Get Help

- Create a Support Ticket via Email
- Virtual Office Hours channel
- UT CMP Developers Microsoft Team
- Sign up for UT CMP Trainings on UTLearn
- Additional UT CMP Support info

SERVICE STATUS

Planned Maintenance (View UT CMP Maintenance Events)

- Disruptive: Mon. - Thurs., 8 PM - Midnight
- Non -Disruptive: Mon. - Thurs., 8 AM - 5 PM

Alerts and Outages

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